

Identifying significant scale-specific spatial boundaries using wavelets and null models: spruce budworm defoliation in Ontario, Canada as a case study

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Abstract We combine wavelet analysis and multiple null models to identify significant spatial scales of pattern and spatial boundaries in historical spruce budworm defoliation in Ontario, Canada. Previous analyses of budworm defoliation in Ontario over the last two outbreaks have suggested three distinct zones of defoliation. We asked the following three questions: (1) is there statistical support for the existence of these three zones? (2) Are the locations of these boundaries consistent between outbreak periods? And (3) how does boundary identification depend on the spatial null model used? Defoliation data for the two outbreak periods (1941–1965 and 1966–2001), and the combined period (1941–2001) were analyzed using a 1D

continuous wavelet transform. Boundaries were identified through comparison of wavelet power spectra of each outbreak period to reference distributions based on three different spatial null models: (1) a complete spatial randomness model, (2) an autoregressive model, and (3) a Gaussian random field model. The Gaussian random field model identified coarser scales of pattern than the autoregressive model. Locally, the Gaussian random field model found significant boundaries similar to those previously described, whereas the autoregressive model only did so for the first outbreak. These results indicate that the coarse scale spatial factors that influenced defoliation were more consistent between outbreaks relative to fine scale factors, and that previously described boundaries were strongly driven by the first outbreak. Wavelet analysis combined with spatial null models provides a powerful tool for identifying non-arbitrary scales of structure and significant spatial boundaries in non-stationary ecological data.

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Introduction

The examination of spatial patterns in ecological data and their spatial scales of expression is important to

better understand spatial processes and to further theoretical development in ecology (Levin 1992; Dungan et al. 2002). Spatial pattern analysis remains challenging when examining large ecological data sets that include patterns determined by multiple processes operating at multiple scales (Holling 1992) such as those created by periodic defoliators like the spruce budworm (*Choristoneura fumiferana* Clem.). When the mean and variance of a data set are not constant over a region being examined, such a pattern may indicate that: (1) the underlying process that generated the data is non-stationary, or (2) more than one process is present and that these multiple processes may be acting at different spatial or temporal scales. Techniques are therefore needed to tease apart different scales of pattern, identify distinct (stationary) sub-regions within larger spatial or temporal contexts, and to assist in making inference about the relative contributions of the different processes to observed spatial patterns.

Identification of ‘non-arbitrary’ scales of ecological pattern is also of particular importance (Wiens 1989). Non-arbitrary scales are those that are relevant to the processes of interest and emergent from data rather than externally imposed by a sampling scheme (Dungan et al. 2002). Statistical significance testing can be used to help identify such non-arbitrary scales that can then be used to better infer scale-specific attributes of ecological processes and to identify regions of local spatial significance. Scale-specific information has been proven useful in assessing wildlife habitat (Morris 2003), understanding natural disturbance dynamics (Turner 1993), and examining spatial population dynamics (Levin 2000).

Methods of scale-specific spatial analysis differ in terms of their conceptual, algorithmic, and mathematical details (Cressie 1993; Borcard and Legendre 2002; Saunders et al. 2002; Bellier et al. 2007). Identification of meaningful and significant spatial scales depends on the data being used and method applied (Csillag and Kabos 2002; Saunders et al. 2002). An important element of any approach that examines non-stationary data is its ability to identify both local and global attributes of spatial pattern. Global attributes summarize an entire study area and local attributes summarize scale-specific information at distinct locations.

Global methods of spatial pattern detection often assume that underlying processes are at least

intrinsically stationary and summarize pattern at a single scale. Such methods include spectral analysis (Keitt 2000; Csillag and Kabos 2002) and variography (Cressie 1993). In contrast, multi-scale methods of pattern analysis can identify both global and local patterns at multiple scales and do not necessarily assume stationarity. Lacunarity analysis (Elkie and Rempel 2001), distance-based eigenvector methods (Borcard and Legendre 2002), and wavelet analysis (Dale and Mah 1998; Csillag and Kabos 2002; Cazelles et al. 2008) are examples of methods capable of identifying global and local scales of pattern. These methods can also be used to identify boundaries in time series or spatial transects (Csillag and Kabos 2002; Fortin and Dale 2005) based on local changes in the scale of spatial pattern.

Wavelets are a particularly useful method of multi-scale spatial pattern analysis that can be used to decompose data into scale-specific components and associate portions of total signal variance with those scales (Csillag and Kabos 2002). Wavelets act similarly to a localized Fourier analysis and have been shown to be more accurate in identifying periodicity than other scale-specific techniques (Saunders et al. 2002). Because of these attributes, wavelet analysis is an increasingly common tool for the analysis of spatial and temporal ecological data (Cazelles et al. 2008). Spatial ecological applications of wavelet analysis include examination of forest canopy gap structure (Bradshaw and Spies 1992), zones of net primary productivity in grasses (Csillag and Kabos 2002), variability in NDVI along a spatial transect (Keitt and Urban 2005), and relationships among boreal forest canopy and understory vegetation (Kembel and Dale 2006).

Here we will use wavelets to study the location of boundaries in spruce budworm (SBW) defoliation in Ontario. SBW is a native defoliator of spruce (*Picea* spp.) and balsam fir (*Abies balsamea*) that affects extensive regions of boreal and mixed-wood forests of eastern Canada (Gray et al. 2000). Defoliation and subsequent tree mortality contributes to the creation of spatial patterns at multiple spatial scales in forest structure that have important ecological and economic consequences (Blais 1983; Gray et al. 2000; Candau and Fleming 2005).

Previous analysis of historical defoliation data in Ontario identified three distinct defoliation “zones” that each contained a central hotspot of defoliation

surrounded by radial gradients of decreasing defoliation (Candau et al. 1998). These “bulls-eye” regions were separated by approximately 100 km wide corridors where defoliation was less frequent centered at approximately 86° and 91° west longitude. However, it is uncertain whether these boundaries represent statistically significant objects (Jacquez et al. 2000), and whether they were in the same locations in both outbreak periods. For the purposes of this research, the term “boundaries” refers to these areas with less than expected cumulative defoliation. Identification of spatial boundaries and their persistence over time from a scale specific perspective can improve our understanding of SBW outbreak dynamics, in particular, by identifying similarities or differences among outbreak periods.

To assess the statistical significance of these boundaries and to examine whether they were consistent between outbreak periods, we analyzed historical spruce budworm defoliation data in Ontario using a continuous wavelet transform (CWT) and compared these results to reference distributions generated using three different spatial null models. We asked three specific questions: (1) Is there statistical support for defoliation boundaries across an east-to west orientation in Ontario as previously described? (2) Are the locations of such boundaries consistent between the two major outbreak periods in the twentieth century? And (3) are there important differences in the ability of different spatial null models to detect these spatial patterns? As a method of spatial pattern analysis, wavelets were chosen over other multi-scale methods because of their ability to identify boundaries in non-stationary data (Csillag and Kabos 2002), ability to assess statistical significance when combined with Monte-Carlo simulation of null models (Torrence and Compo 1998), and demonstrated strengths in accurate scale detection (Saunders et al. 2002).

Methods

Wavelet analysis

Wavelet analysis can be applied to time series, spatial transects, or two dimensional images to identify patterns at different scales or frequencies (Daubechies 1992; Dale and Mah 1998; Cazelles et al.

2008). Wavelet analysis involves successive passes of a wavelet template of increasing size over a data series and assesses the degree of similarity between the pattern of the data and the shape of the template at each size and location. The values that describe this degree of fit are referred to as wavelet coefficients. At a given location and template size, the value of the wavelet coefficient is high where the pattern of the data matches the shape of the template, and close to zero when it does not (Bradshaw and Spies 1992; Dale and Mah 1998). As this analysis is repeated at various template sizes, the continuous wavelet transform decomposes a data series into multiple scales and creates new data series of wavelet coefficients for each level of decomposition. The set of coefficients at each scale reflects the local variation in the original series at that scale.

A plot of wavelet coefficients in location by scale space is referred to as a wavelet power spectrum (e.g., Figs. 1g, h, i). To determine scale-specific information independent of location, we examined wavelet variance at each scale. Wavelet variance is the sum of the squared wavelet coefficients at a particular scale weighted by the number of coefficients (Bradshaw and Spies 1992; Percival 1995). Scales associated with high wavelet variance are considered scales of interest (Bradshaw and Spies 1992). A plot of wavelet variance versus scale is called a scalogram (Dale and Mah 1998) or a global wavelet spectrum (GWS; Percival and Walden 2000). Wavelet analysis can be considered a scale-specific approach to variance decomposition. For this analysis, we used a continuous second order derivative of Gaussian (DOG) wavelet, also referred to as the “Mexican Hat” wavelet, to analyze the cumulative defoliation data for Ontario. Detailed descriptions of the wavelet methods used are presented in Torrence and Compo (1998).

Defoliation data

Aerial surveys of budworm defoliation in Ontario have been conducted annually since 1941 by the Forest Insect and Disease Survey (FIDS) of the Canadian Forest Service (CFS). Defoliation mapping is done from an aircraft and involves sketching tree discoloration caused by feeding budworm larvae (Howse 1995). Three levels of defoliation are defined on the basis of proportion of foliage damaged: light

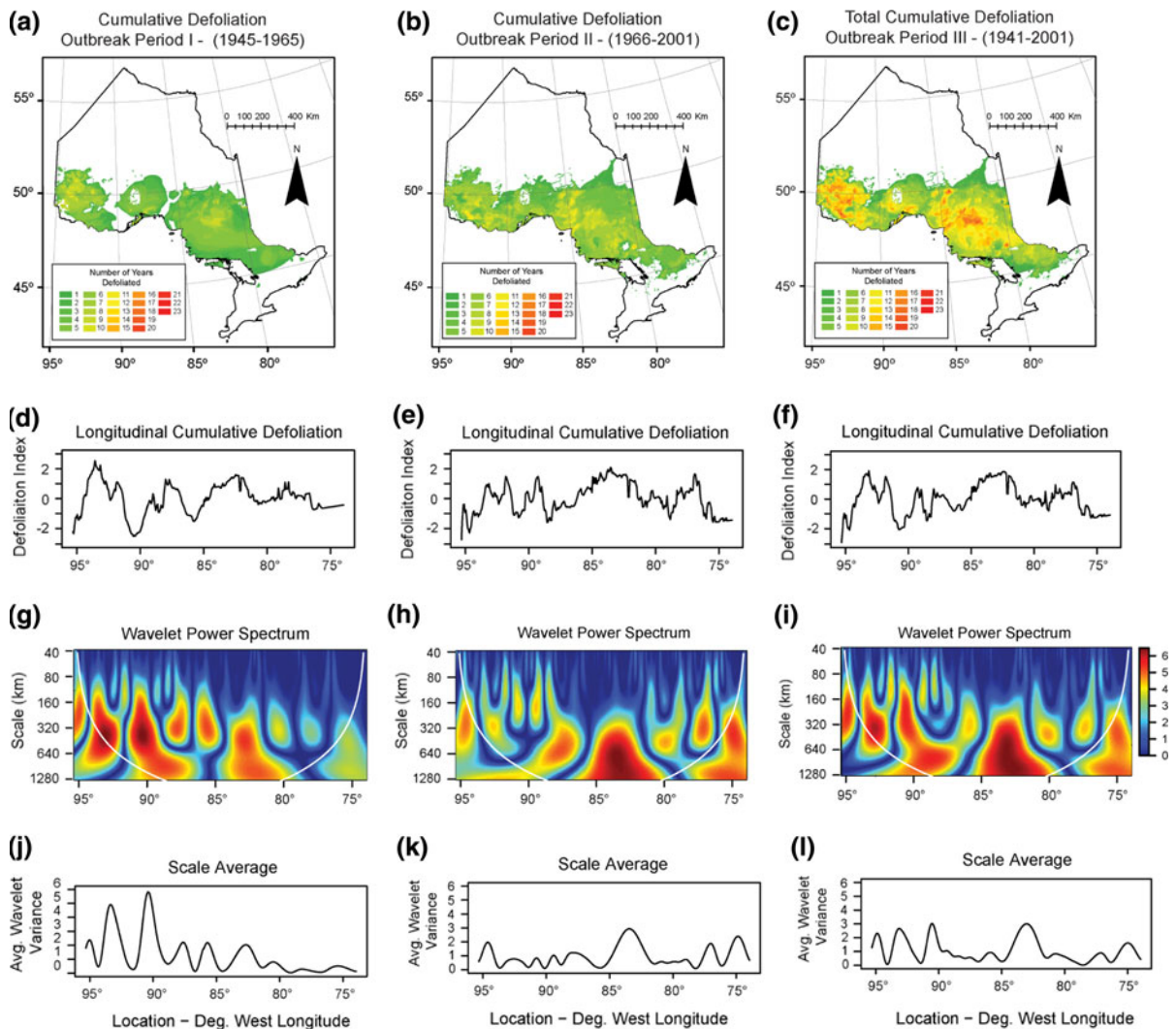


Fig. 1 Cumulative defoliation in Ontario for (a) outbreak period I (1941–1965), (b) period II (1966–2001), and (c) the combined duration, outbreak period III (1941–2001). Maps represent the total number of years classified as “moderate” or “severe”. Cumulative defoliation represented as longitudinal transects in 5 km wide increments (d, e, f). Wavelet power spectra for each outbreak period based on a second order DOG wavelet (g, h, i). Warmer colors represent regions in scale-

location space with large wavelet coefficients. Regions below the *white curve* are unreliable due to boundary effects. Scale-averaged wavelet variance from east to west across the defoliation transect for each outbreak period (j, k, l). These values represent the average values of the global wavelet spectra summarized over all scales for each location at scales larger than 40 km (j, k, l). See text for further details

(0–25%), moderate (26–75%), and severe (76–100%).

There were two major outbreaks in the last century for which defoliation data were recorded (Candau et al. 1998). Outbreak period I is defined as 1941–1965 (24 years; Fig. 1a), and outbreak period II as 1966–2001 (35 years; Fig. 1b). A third period was also analyzed that includes all years (1941–2001;

Fig. 1c) and is referred to as outbreak period III. Rasterized defoliation maps were used to calculate the total number of years defoliated in the moderate or severe category for both outbreak periods in 5×5 km pixels. Raster maps were summarized by column to produce three transects of cumulative defoliation in 5 km longitudinal increments from east to west. Data were reduced to a linear transect to

develop our wavelet testing procedure and to make our results comparable to the boundaries and longitudinally divided zones in Ontario described by Candau et al. (1998). Indeed, previous analyses of defoliation in Ontario have identified a dominant east–west orientation (Candau et al. 1998; Candau and Fleming 2005). Cumulative defoliation at each location was standardized by number of rows (i.e., latitudinal range) and number of years within the recorded outbreak period. Cumulative defoliation transects for outbreak periods I, II, and III are shown in Fig. 1d, e, and f, respectively. All data were detrended prior to model parameter estimation and wavelet analysis.

Candau et al. (1998) identified three separate regions of defoliation in Ontario using these defoliation data. Further analysis revealed that patterns of defoliation are linked at broad scales to climatic conditions whereas finer scale patterns are influenced by patterns of host tree availability (Candau and Fleming 2005). Bellier et al. (2007) also used these data to compare two scale detecting statistical methods and to investigate the spatial relationships between SBW defoliation and several bioclimatic variables. They found support for the scale dependent relationships between defoliation and bioclimatic variables described by Candau and Fleming (2005). These previous studies provide a solid description of Ontario's historical spruce budworm defoliation record upon which we expand using the wavelet significance testing method described below.

Null models

Spatial significance testing using null models involves comparing a statistic derived from data of interest to a distribution of that statistic generated by a spatial null model that represents a spatial process in the absence of the assumed pattern generating phenomenon (Jacquez et al. 2000). Using such an approach, Torrence and Compo (1998) introduced a framework for using normalized wavelets and wavelet variance to identify scale-specific statistical significance in time series. This significance testing was based on a comparison of a spectral decomposition of a time series to the expected spectral density of red and white noise null models using a Chi-square test.

This significance testing framework was applied to identify significant scales of spatial pattern and local

boundaries in budworm defoliation. However, because the components of the Fourier transform of the original defoliation data series are not normally distributed, the Chi-square distribution could not be used as a reference. Instead, Monte Carlo simulations were used to generate reference distributions of three null models that differ in terms of their auto-covariance structure. The models used were (1) a complete spatial randomness model (CSR; Fortin and Dale 2005), (2) a second-order autoregressive model (AR2; Bailey and Gatrell 1995), and (3) a stationary Gaussian random field model (GRF; Diggle and Ribeiro 2006).

The AR2 model is best suited for temporal (i.e., directional) data and was applied to identify boundaries in the directional defoliation transect in which spatial dependencies can depend on locations on either side of a focal location (i.e., symmetrical dependency). However, previous work on SBW defoliation in Ontario has shown a strong directional trend in their tendency to travel from east to west (Candau et al. 1998; Candau and Fleming 2005), making an asymmetrical dependency possibly more likely than an equally weighted symmetrical dependency. The ability of this relatively simple model to identify boundaries was compared to the more complex explicitly spatial (i.e., symmetrical) GRF model.

Null model parameters were estimated using maximum likelihood for each defoliation period and used to simulate 9,999 realizations of each model (Table 1). Prior to wavelet analysis, all defoliation transects were symmetrically padded with zeros to make their length dyadic (i.e., to the next largest power of two). Such dyadic padding is a technical requirement of wavelet analysis. All simulated series were similarly padded and standardized to a mean of zero and unit variance. A continuous wavelet transform was performed on each padded and standardized null realization to generate a distribution of wavelet power spectra to which the equivalent spectra of the three cumulative defoliation transects were compared. That is, the wavelet based scale-specific decompositions (i.e., spectra) of the original defoliation data were compared to a set of similar decompositions of each null model. Situations where the defoliation spectra exceed the 95th percentile of the distribution of simulated global or local wavelet spectra represent significant scales or locations, respectively. The null models used are summarized below.

Table 1 Null model parameter estimates for autoregressive (AR2) and Gaussian random field (GRF) models

Outbreak Period	AR2 Parameters				GRF Parameters			
	α_1	α_2	<i>SE</i>	<i>AIC</i>	τ	σ^2	ϕ	<i>AIC</i>
I	1.238	−0.265	0.043	−971.80	0	0.879	2.459	−882.65
II	1.180	−0.231	0.084	−760.60	0	0.086	1.833	−697.44
III	1.160	−0.205	0.073	−810.00	0	0.275	2.320	−403.03

α_1 and α_2 represent autocorrelation coefficients for the AR2 model and *SE* represents the standard error of these estimates. For the GRF model, τ represents nugget variance, σ^2 represents the sill, and ϕ represents the range estimate. Parameters for both models were estimated from detrended data using maximum likelihood. For both sets of parameters, *AIC* represents the Akaike Information Criterion

Complete spatial randomness

The simplest null model involved random resampling of the complete set of the original data transects without replacement (Fortin and Dale 2005). This model assumes that there is no spatial structure in the defoliation data and tests the alternative of any spatial structure.

Autoregressive model

A common null model for natural and geophysical series is red noise (Halley 1996; Torrence and Compo 1998). A determining characteristic of red noise is that the amount of signal variance decreases as frequency, or scale, becomes coarser (Rodionov 2006). Red noise represents an intrinsically stationary autocorrelated signal that integrates previous events and can be used to represent system memory or inertia (Halley 1996). Because of its tendency to integrate previous events, red noise often exhibits long intervals during which a data series remains above or below its mean value (Rodionov 2006). As such, a red noise null model is well suited for the analysis of the cumulative defoliation data and was able to generate periodicities similar to those observed in the data (Figs. 1d, f).

Red noise was simulated using a second-order autoregressive model (AR2) because it represented the lowest order (i.e., simplest) model for which the AIC value of the maximum likelihood fitting procedure indicated the best fit for each data set (data not shown). Parameters were estimated using the ‘*FitAR*’ function in R (McLeod and Zhang 2008; R Development Core Team 2008). Use of this red noise null model tests the null hypothesis that the spatial structure of historical defoliation in Ontario is no

different than that generated through the lag-2 autoregressive model. Failure to reject this null hypothesis would indicate that there are no significant boundaries or regions of non-stationarity relative to the null model in the defoliation record. The linear second-order autoregressive (AR2) model was specified as:

$$Y_i = \alpha_1 Y_{i-1} + \alpha_2 Y_{i-2} + Z_i, \quad (1)$$

where Y_i is the value at location i , α_1 and α_2 are the estimated autocorrelation coefficients (Table 1), and Z_i represents a Gaussian random deviate that is independent and identically distributed with a mean equal to zero and unit variance. Y_0 was set to 0.

Gaussian random field model

Spatial models often require distinction from temporal models due to the simultaneous dependency on multiple locations (Fortin and Dale 2005). Variograms have been used extensively in the field of geostatistics and ecology to analyse spatial patterns (Cressie 1993; Fajardo and McIntire 2007). Geostatistical models can also be used to simulate spatial patterns using a chosen variogram model, process model, and parameter estimates (Goovaerts and Jacquez 2005).

We used a Gaussian stochastic process based exponential variogram null model. Gaussian stochastic processes are widely used for geostatistical simulation because of their availability, tractability and relative simplicity (Diggle and Ribeiro 2006). The resulting simulated data are referred to as a Gaussian Random Field (GRF). Variogram-based parameter estimates for GRF simulation are shown in Table 1. Parameters were estimated using maximum likelihood in the *geoR* package (Ribeiro and Diggle

2001) and simulations were performed using the *RandomFields* package for R (Schlather 2009). The GRF null model tests the null hypothesis that the spatial structure of historical defoliation in Ontario is no different than that generated by this variogram-based model. Failure to reject this null hypothesis would also indicate that there are no significant boundaries or regions of non-stationarity in defoliation relative to the null model. The simulated value of a GRF model at a location x_i is defined as:

$$Y_i = S(x_i) + Z_i, \quad (2)$$

where S represents a spatial stochastic process specified completely by its mean and covariance structure (Diggle and Ribeiro 2006). Z_i represents a Gaussian random deviate that is independent and identically distributed with a mean equal to zero and variance equal to the nugget variance, τ^2 . Here, S is a stationary process that follows a multi-variate Gaussian distribution with variogram function C . The variogram function describes the covariance between two points (x_i, x_j) separated by lag distance h and can be written as:

$$C(h) = \sigma^2 * \rho(h), \quad (3)$$

where σ^2 is the variogram variance parameter (i.e., the sill) and $\rho(h)$ represents the correlation function, parameterized by lag distance h and the range parameter, ϕ . The correlation function for the exponential variogram model is:

$$\rho(h) = e^{(-h/\phi)}. \quad (4)$$

Significance testing

Globally significant scales of pattern were determined through comparison of the global wavelet spectrum of the defoliation transects and the 95th percentile of each null model's distribution of spectra and is equivalent to assessing significance at the level of $\alpha = 0.05$. Global assessments of significance test the null hypothesis that the scale-specific spatial structure in defoliation can be completely described by the null model. Scales at which the defoliation data's global spectrum had higher wavelet variance than the 95th percentile from the simulated null distribution were considered significant.

Local significance testing used sets of local power spectrum coefficients to identify boundaries in cumulative defoliation data and tests the null hypothesis

that the entire study region can be uniformly described by a single spatial null model. To do so, local scale-averaged wavelet power was calculated using only significant scales determined from the original wavelet power spectrum (e.g., Fig. 1d). That is, we calculated the average wavelet variance at all locations restricted to significant scales identified above (Torrence and Compo 1998). Scale-averaged wavelet power describes location specific variations in wavelet variance in contrast to the wavelet variances that are averaged over all locations for each scale that comprise a scalogram. Scale-averaged wavelet variances for each location were calculated for each of the 9,999 simulated realizations. All local wavelet variances for all simulations (i.e., $9,999 \times 314$ locations) were sorted in increasing order. Actual locations where the scale-averaged wavelet variance from the defoliation data exceed the 95th percentile from the sorted distribution of all simulated values indicate locally significant ($P < 0.05$) regions, or peaks, in defoliation relative to the selected null model. That is, areas that stray outside of the range of expectation given the model and estimated parameters. Regions in between such significant peaks represent significant spatial boundaries in defoliation that were visually compared to the expected (i.e., previously described) defoliation boundaries.

Results

Defoliation data

The defoliation profiles for each outbreak period reveal different patterns in cumulative defoliation (Figs. 1a, b). The areas identified by Candau et al. (1998) as zones of disproportionately low defoliation are clearly defined in the first outbreak around 91° and 86° west longitude, but are less discernable in the second. Outbreak period II shows a strong peak of defoliation in the centre of the province at approximately 85° west longitude. The combined outbreak period (Fig. 1c) shows aspects of both outbreaks but appears dominated by the first.

Parameter estimates indicate that, in general, the AR2 model fit the data better than did the GRF model for each outbreak period (i.e., had lower AIC values; Table 1).

Wavelet power spectra

Wavelet power spectra show that variability in cumulative defoliation is greater at coarser scales. This pattern is indicated by the warmer colored areas associated with larger template size/scale (Figs. 1g, h, i). Global assessments of the scale-specific properties of the decomposed signal independent of location were achieved by summarizing the mean of the squared wavelet coefficients across all locations to produce a scalogram, or global wavelet spectrum (Fig. 2). Local, scale-averaged assessments of wavelet variance are achieved by summarizing the squared wavelet coefficients over all scales (i.e., scale-averaged wavelet power; Figs. 1j, k, l).

Global wavelet spectra (GWS; Fig. 2) show large amounts of variance at coarser spatial scales and little

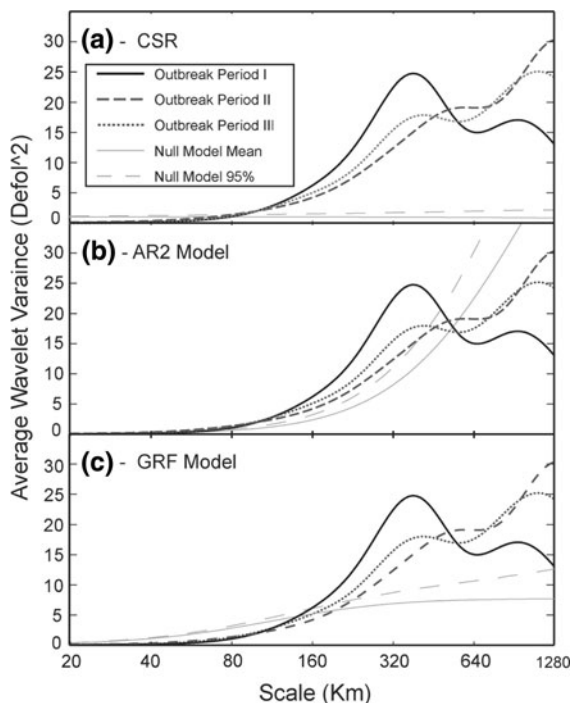


Fig. 2 Global wavelet spectra, or scalograms, for each outbreak period compared to the mean and 95th percentile spectra for each null model: (a) complete spatial randomness (CSR), (b) autoregressive lag-2 model (AR2), and (c) linear, stationary Gaussian random field model (GRF). Regions where the scalograms for each outbreak exceed the 95th percentile curve indicate significant scales of spatial structure. Mean and 95th percentile values were calculated from 9,999 realizations of each null model

fine scale variability within each outbreak period, which indicates that the null models capture much of the fine scale variability in the defoliation data. Outbreak I had a distinct peak in variance at intermediate-large scales centered at a window of approximately 320 km, Outbreak II showed a similar change in variance with increases in scale but showed a peak around 500 km, and continued to increase with further increases in wavelet template size.

The scale-averaged wavelet power shows that local patterns of wavelet variance differed between outbreak periods I and II (Figs. 1j, k). More peaks in variance are present in outbreak II, however, stronger peaks are present in outbreak I, particularly in the west. The strong central peak in outbreak II is also apparent in the local wavelet variance profile (Fig. 1h). Examination of the combined defoliation profile reveals that the western influence of outbreak I is particularly strong, although the middle peak from outbreak II is also present.

Significance testing

When comparing defoliation to the CSR null model, all scales greater than 80 km in the outbreak data were significant (Fig. 2a). The scalogram for the AR2 model (Fig. 2b) showed an exponential increase in variance captured with increasing scales. Outbreak periods I and II exceeded the 95th percentile of the AR2 model for intermediate scales (e.g., 80–450 km) but were within the range of expectation at scales greater than approximately 450 km. Where the scalogram for period I exceeded the null model it did so clearly, whereas the scalogram for outbreak II only marginally exceeded the null model at intermediate scales.

In contrast to the AR2 model, the scalogram for the Gaussian random field model (GRF) showed a sigmoidal increase in wavelet variance with increasing scales (Fig. 2c). Both outbreaks exceeded the range of expectation at scales greater than approximately 160 km, indicating global significance in scale of pattern in the cumulative defoliation data for all intermediate and coarse scales.

Local scale-specific significance was identified for each null model relative to the 95th percentile of wavelet variance over all simulations and locations, restricted to the globally significant scales identified above (Fig. 2). Relative to the CSR model, nearly all

locations were significant (Fig. 3a). Regions of local significance relative to the AR2 and GRF models are shown as peaks that exceed the 95th percentile (dashed horizontal line) in Figs. 3b and c, respectively. Regions between significant peaks in local wavelet variance represent boundaries as do the significant troughs identified by the AR2 model. Significant regions across the defoliation transect are summarized in Fig. 4. Two-dimensional representations of these regions of mapped onto the province of Ontario are shown in Fig. 5.

When comparing outbreak period I to the AR2 model, support was found for six significant regions, of which three were peaks and three were troughs. The significant peaks, that is, regions with greater than expected cumulative defoliation relative to the null distribution, were (1) 94–92.5°, (2) 88.5–87.5°, and (3) 84–82° west longitude. The significant troughs, that is, regions with lower than expected cumulative defoliation were found at (1) 95.5–95°, (2) 91.5–89.5°, and (3) 86.5–85.5° west longitude (Figs. 3b(i), 4a). Four significant regions were found in outbreak period II (Figs. 3b(ii), 4b). One very narrow trough was identified from 95–94.5° west longitude. Two significant peaks were identified between 85 and 82.5° and 77.5–77°, and a final trough from 75.5 to 74° west longitude. Comparison

of outbreak period III with the AR2 model revealed five significant regions, three of which were peaks, and two of which were troughs (Figs. 3b(iii), 4c). The locations of the significant peaks were (1) 93.5–92°, and (2) 84.5–82° west longitude. Troughs were located between (1) 95.5–95°, (2) 91–90°, and (3) 75.5–74.5° west longitude. Note that the importance of the far western boundaries identified is uncertain due to potential edge effects as a result of the dyadic padding process.

When comparing outbreak period I to the GRF model, support was found for four significant regions: (1) 95–93°, (2) 91–87°, (3) 85.5–80.5°, and (4) 75.5–74.5° west longitude (Figs. 3c(i), 4a). These regions were very wide and did not correspond with the fine scale peaks and troughs identified by the AR2 model. Outbreak period II also contained four significant regions that were larger than those identified in outbreak period I: (1) 95.5–93.5°, (2) 92.5–87°, (3) 86.5–80°, and (4) 78.5–74° west longitude (Figs. 3c(ii), 4b). Boundaries were found at similar, but not identical locations in the two outbreak periods. Comparison of the GRF model to outbreak period III identified three significant regions that were very similar to those identified in period II. These regions were found at (1) 91.5–87°, (2) 85.5–80°, and (3) 77.5–74° west longitude (Figs. 3c(iii) and 4c). The

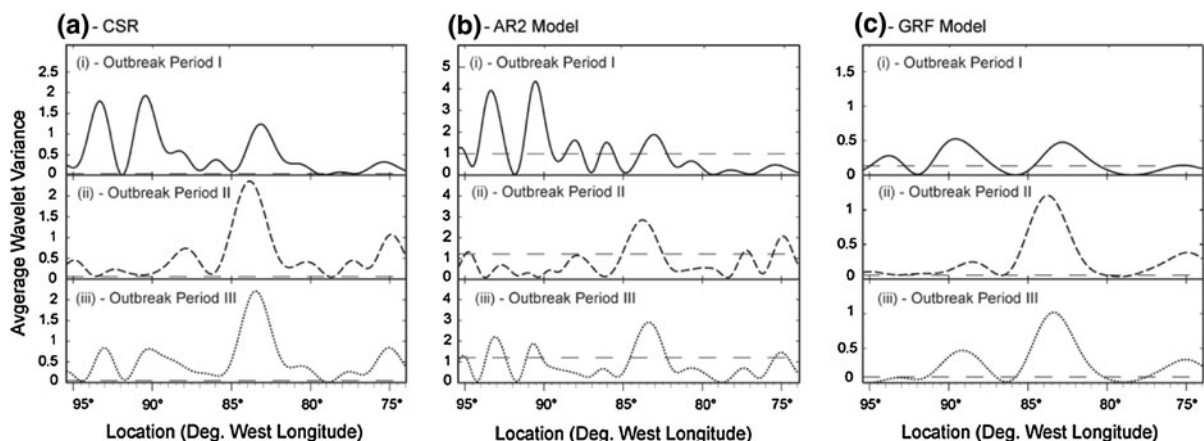


Fig. 3 Local scale averaged wavelet variance for each outbreak period (i–iii) relative to three null models (a–c). Dashed lines represent the average 95th percentile of the scale-averaged wavelet variance at each location over all simulated replicates restricted to scales identified as significant and

illustrated in Fig. 2. Solid lines represent the scale-averaged variance of cumulative defoliation restricted to significant scales. Regions where the solid line exceeds the dashed lines indicate significant regions of local scale-specific significance

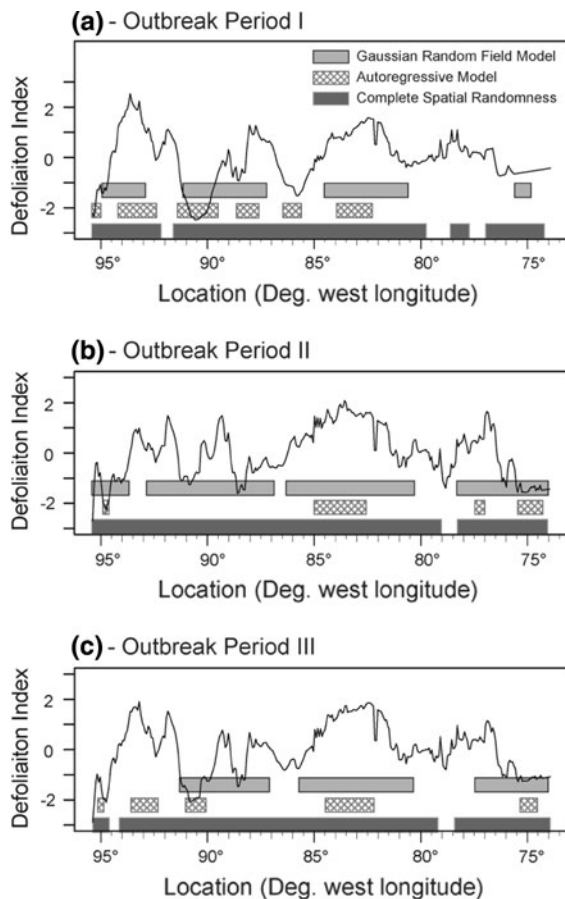


Fig. 4 Summary of significant regions in cumulative defoliation and local boundaries. *Black lines* represent cumulative defoliation shown in Figs. 1d, e, and f. *Filled horizontal bars* identify regions with locally significant amounts of historical defoliation relative to the three null models. Intervening non-significant regions represent boundaries. In the autoregressive lag-2 model (AR2), some significant regions correspond to peaks in the defoliation record, and others correspond to troughs, i.e., regions with very little defoliation, that also represent spatial boundaries

boundaries separating significant regions in outbreak period III overlap with those found in periods I and II.

Discussion

Although many methods exist for identifying scales of pattern in ecological data, few test specifically for scale-specific boundaries. We described the steps required to perform parametric bootstrap significance testing of observed global scales of pattern (i.e., GWS), and local spatial boundaries using existing

wavelet analysis algorithms (Torrence and Compo 1998). Novel aspects of this research are (1) application of wavelets to identify local boundaries in spruce budworm defoliation data independent of global scale-specific significance, and (2) the comparison of several null models with varying auto-covariance structures to identify boundaries in spatial pattern.

Spatial boundaries

Wavelet-based comparison to a simulated AR2 model found statistical support for the existence of previously described defoliation boundaries in outbreak period I and the combined period (outbreak period III), but not period II. The GRF model found strong support in outbreak period I and some support in periods II and III. Parameter estimates indicate that, in general, the AR2 model fit the data better than did the GRF model for each outbreak period (i.e., had lower AIC values; Table 1). However, the degree of fit of the model to the data as measured by AIC does not bear on the consistency of the spatial boundaries and their similarity to those previously described.

Many of the significant regions identified through boundary delineation correspond with obvious peaks (AR2 and GRF) and troughs (AR2 only) in the original cumulative defoliation profiles. The GRF model did not identify troughs because of the coarser scales of pattern it retains in the defoliation data due to its sigmoid-shaped scalogram (Fig. 2c). In contrast, the AR2 model was able to identify troughs because the very coarse scales were filtered out of the defoliation data because of its exponential scalogram. Different null models contain different assumptions about underlying processes and as such vary in their ability to assess statistical significance and identify boundaries at different scales. From a methodological perspective, the identification of obvious peaks and troughs is reassuring because it indicates that the wavelet testing method we presented can identify expected discontinuities in relatively simple linear data and can also likely identify discontinuities in more complex data.

Consistency between outbreak periods

Boundaries among local regions were dependent on the outbreak period examined and the null model

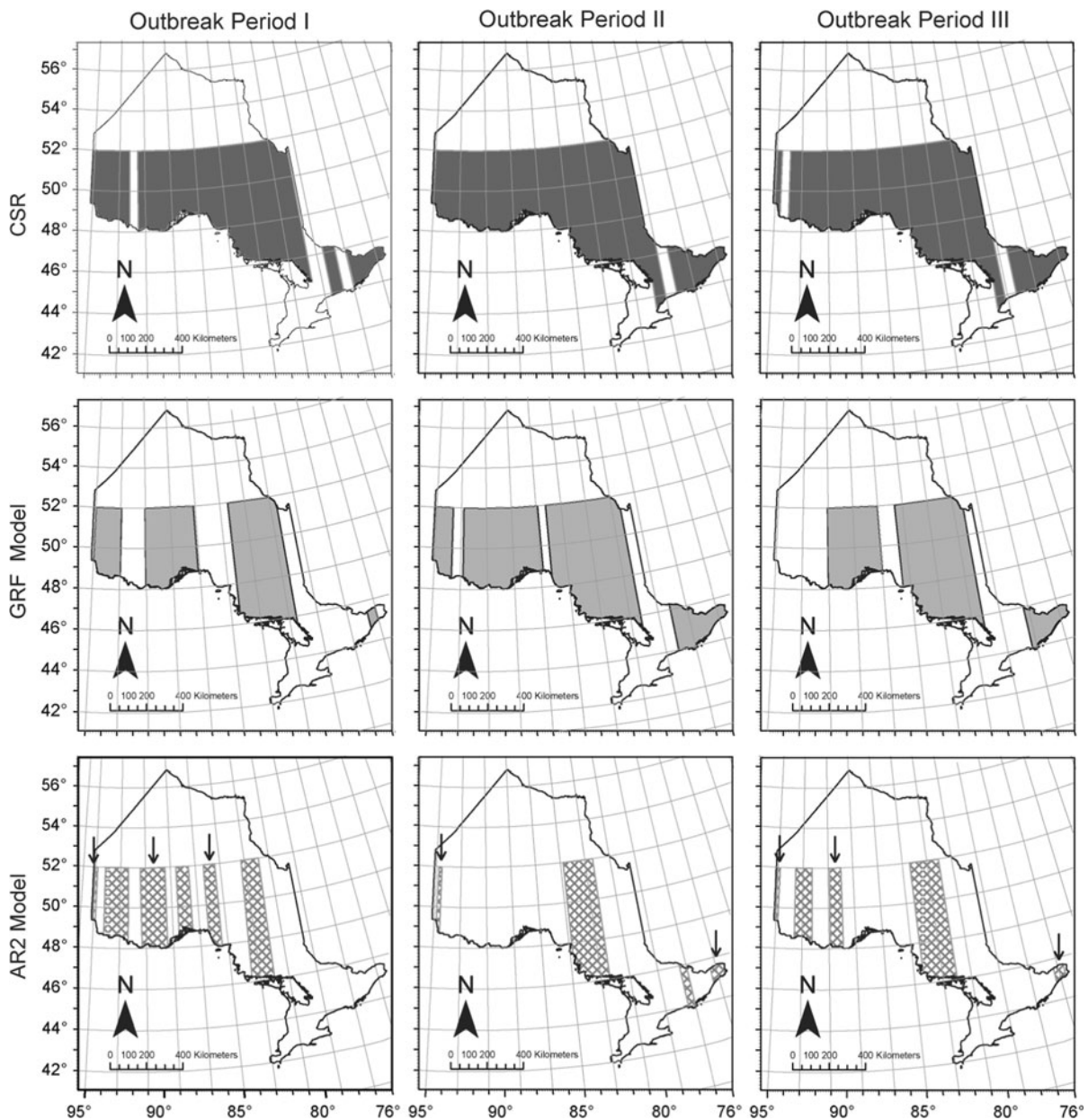


Fig. 5 Summary of significant regions shown in Fig. 4 plotted onto a map of Ontario relative to the three null models: (1) complete spatial randomness model (CSR); (2) autoregressive lag-2 model (AR2); and (3) Gaussian random field model (GRF). Boundaries are implicit between significant regions. Filled areas indicate regions with locally significant amounts

of historical defoliation relative to the three null models. In maps showing AR2 results, downward pointing arrows indicate significant troughs, that is, regions with less than expected cumulative defoliation. All other filled areas represent significant peaks in cumulative defoliation

used. The CSR null model did not distinguish among regions, which highlights the importance of using autocorrelated null models when examining spatial data (Fortin and Jacquez 2000). In addition, null models that assume spatial independence tend to

result in false positives (Goovaerts and Jacquez 2005).

The AR2 null model identified significant troughs in outbreak period I that correspond with previously described boundaries centered at approximately 86°

and 91° west longitude. In outbreak period II, the AR2 model only found significance in a large central eastern peak that was also identified in outbreak period I. Outbreak period III (i.e., combined periods I and II) showed the same western zone of low defoliation around 91°, but the unclear signal from period II appears to have washed out the division between the central and eastern regions. From the perspective of the AR2 null model, the zones described by Candau et al. (1998) may only represent meaningful divisions of the province and its defoliation history with respect to outbreak period I. However, because surveys captured only the latter portion of the first outbreak period (this outbreak prompted monitoring), these data may be biased to identify stronger patterns in the first outbreak, given that we standardized each location by number of years in the outbreak.

The GRF model did not identify significant troughs because troughs are present at a finer spatial scale than the GRF model is able to detect (Fig. 2c). GRF boundaries were thus interpreted from the non-significant regions between significant regions and suggests that the higher than expected defoliation zones are perhaps of more interest than the boundaries between such zones. Each outbreak period contained three significant regions (Fig. 4), and outbreak period II contained an additional region in the far west of the study area (Fig. 4b). Generally, boundaries between regions were consistent among the three outbreak periods, although boundaries in period I and III were broader (i.e., spanned more degrees of longitude) than those in period II (Figs. 4, 5). The GRF model was also able to identify distinct regions in the west of the study area in outbreak period II where the AR2 model could not (Fig. 4b). The boundaries shown in outbreak period III are most similar to those in period I (Fig. 4), further suggesting that the strong patterns in period I are dominant in the combined outbreak period, and there are differences to consider in the spatial factors that determined defoliation patterns between outbreak periods I and II. However, relative to the AR2 model, the boundaries interpreted from the GRF null model more strongly suggest that the zones described by Candau et al. (1998) represent meaningful divisions of the province and its defoliation history.

Spatial null models

Ecological data are often structured at multiple scales, which can make it difficult to assess statistical significance. Wavelet analysis is well suited to this challenge but significance still depends of the type and complexity of the null model used (Jacquez et al. 2000). In this study we used null models that were specified as a single spatial process. Nested variogram models (Wackernagel 2003; Bellier et al. 2007) and spatial “surrogate approaches” (Rouyer et al. 2008) can be used to include multiple processes in a null model to assess significance relative to multiple types of spatial structure. However, complex null models risk including too much spatial structure; over-fitting the null model can make it unlikely that the data of interest will be significantly different, whereas underfitting can result in the opposite. For complex ecological data where it is not always clear what type of spatial structure should be included, multiple null models with different spatial structures should be compared. Selection of a reference distribution is an important step to scale-specific assessments of significance and boundary detection and it should be kept in mind that some models are more “null” than others.

Differences between the regions identified by the GRF and AR2 models likely stem from different scales of specialization of the two models (Fig. 2). Differences in fine scale patterns among outbreak periods as identified through comparison to the AR2 model suggest that fine scale spatial factors were not consistent in each outbreak period. The coarse scale analysis provided by comparison to the GRF model indicates a similar effect but to a lesser degree, suggesting that coarse scale factors were more consistent over the two outbreak periods.

Biologically, the significant scales of pattern and local significant regions could reflect the scale of host forest patchiness or other environmental factors such as climate or additional spatial population dynamics (e.g., dispersal). Blais (1983) suggested that the three defoliation regions correspond to areas with a high density of mature host species (i.e., spruce and fir), the centers of which acted as epicenters to the developing epidemic. Candau and Fleming (2005) found the winter and spring temperatures played an important role in determining defoliation patterns and

that at coarse scales climate most strongly predicted defoliation but suggested that finer scales may be driven by local patterns in host availability. At finer scales (20–50 km), there are indications that the contagious aspects of SBW population dynamics may also dictate where defoliation occurs (Candau and Fleming 2005). Local dispersal processes combined with variable patterns in forest composition could account for the temporal non-stationarity in defoliation boundaries identified between the two outbreak periods.

The spatial legacy of the first outbreak period could also account for the different spatial patterns detected in the second outbreak. That is, patterns of stand mortality and successional consequences from the first outbreak period could have created patchy forest conditions that limited the synchronicity and continuous extent of budworm damage relative to the first outbreak. However, in the context of this study we are unable to fully determine whether this was the case. Indeed, it is uncertain how to fully disentangle the roles of forest structure, climate, and predators on observed outbreak patterns. Nonetheless, the long term dynamic interactions among forest disturbances may play an important role (McCullough et al. 1998) and in the context of uncertain landscape-scale successional dynamics (Schroeder and Perera 2002), is a area that deserves further investigation. This is especially true when considering the uncertain consequences of human management systems (e.g., logging) on natural disturbances processes (Cooke et al. 2007).

We used the DOG wavelet to investigate scale and structure of cumulative SBW defoliation in Ontario. The DOG wavelet is a commonly used real-valued template for wavelet analysis (Kembel and Dale 2006, Cazelles et al. 2008). Selection of wavelet templates necessarily involves trade-offs in terms of a wavelet's spatial (or temporal) resolution and frequency resolution (Daubechies 1992). For example, the DOG wavelet is well localized in space (or time) but poorly localized in the frequency domain, whereas the inverse is true for the Morlet wavelet (Mi et al. 2005, Cazelles et al. 2008). Because we were specifically interested in identifying local spatial boundaries, localization in space is of greater relative importance and the DOG wavelet was selected. Furthermore, it was not necessary to employ a complex template (e.g., Morlet) because we are not

interested in the distinction between phase and amplitude of the wavelet coefficients.

Other methods of wavelet based significance testing compare an empirical wavelet power spectrum to that of a simulated null distribution (Cazelles et al. 2007, 2008; Rouyer et al. 2008). In these analyses, significance is assessed simultaneously in scale (frequency) and location (spatial or temporal) space rather than separately. In contrast, the two-step significance testing procedure presented here filters out non-significant scales of pattern prior to identifying locally significant boundaries. The strength of this approach is that wavelets are used as both a band-pass filter (Bruce and Gao 1996) to remove “noise”, defined as non-significant scales, and also to identify local variations in scale-averaged wavelet variance. However, a weakness of this approach is that it is possible to filter out global scales of structure that contain important local information. Significance testing through comparison to the full local wavelet power spectrum contains all locations and all frequencies, but interpretation is potentially confounded by the inclusion of non-relevant scales. The relative utility of these approaches will depend on the extent to which the research questions focus on localization in space (or time) and localization in frequency. Localization in frequency is often more meaningful in time series analysis for which periodicity is the central research objective, than it is for spatial analysis. As such, we suggest that the two step significance testing method that includes de-noising, or removing of non-significant scales of pattern, may be of more use in spatial studies. However, choice of analysis method ultimately depends on specific research objectives and data characteristics.

Although wavelet analysis is powerful for identifying scale specific information, there are several challenges to using this approach that should be kept in mind. The first relates to boundary effects. As the wavelet template increases in size, it can begin to overlap the edges of the series being analyzed. As such, information derived near the edges of series at coarse scales should be interpreted cautiously. The second challenge relates to selection of a wavelet template. Different templates can reveal different structure, and as discussed above, necessarily involve a trade-off in the ability to localize information in the spatial or frequency domain. Multiple wavelet templates should be explored in initial stages of data

analysis (Daubechies 1992). The full consequences of different templates are yet to be fully investigated in ecological data. Wavelet analyses are also not applicable in all contexts because they are generally restricted to regular sampling (but see: Rosenberg 2004). Finally, wavelets can be performed on discrete, dyadic bases, or in a continuous context. Choice of the discrete or continuous forms will depend on whether one desires the extracted sets of wavelet coefficients to be independent. Here, we used a continuous wavelet transform to more fully capture the ranges of scale of pattern in cumulative defoliation, rather than extract independent scale specific series to be used in a predictive modeling framework (e.g., Keitt and Urban 2005).

Conclusions

We demonstrated a wavelet based approach in combination with null models to identify statistically significant scales of pattern and local spatial boundaries in historical defoliation data. With respect to local patterns of spruce budworm defoliation we found some support for the coarse scaled defoliation zones previously described by Candau et al. (1998) and identify additional variability between outbreak periods at finer spatial scales. Further analyses of these data in either a 1D or 2D context would benefit from examining the different outbreak periods separately. Support for the locations of significant local boundaries was strongly driven by the spatial patterns present in outbreak period I, and indicated some degree of temporal non-stationarity among outbreak periods. Further investigation into the relationships among defoliation, forest structure, climate, and other environmental factors in the different outbreak periods in a scale specific context will be useful to further explain these differences in spatial pattern.

As issues pertaining to scale continue to be important in ecology (Levin 1992), wavelets should become an increasingly applied method of analysis due to their indifference to non-stationarity, ability to summarize both local and global patterns, and the availability of a well developed mathematical framework, algorithms, and software. Wavelets represent a powerful method of analysis that we hope will assume an increasingly important role in the ecologist's spatial and temporal analysis toolbox. Many

opportunities exist for further development of wavelet analysis in ecology, including application of these methods to multivariate data and to two dimensional spatial data.

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